

CyberInvestigator

AI Fraud Network Intelligence Platform

“See the fraud network, not just the transaction.”

Theme: AI for Digital Public Safety — Defeating Counterfeiting, Fraud & Digital Arrest Scams

Submitted for ET AI Hackathon 2.0

Executive Summary

CyberInvestigator is an AI investigation platform built for law-enforcement and financial-crime units. It ingests transaction and complaint data, detects coordinated fraud rings — not just individually suspicious transactions — identifies the ringleader, traces the money trail, maps cross-jurisdiction spread, answers investigator questions through a tool-calling AI agent, and generates tamper-evident, court-admissible intelligence packages. The platform is designed to shift financial-crime response from reactive case investigation to proactive network detection before mass victimisation occurs.

Problem Statement

- India registered **1.14 million** cybercrime complaints in 2023 — up **60% year over year** (Ministry of Home Affairs).
- **Digital arrest scams** defrauded citizens of over ₹**1,776 crore** in just the first nine months of 2024 (Ministry of Home Affairs).
- The RBI's Annual Report 2025 flagged **record high-quality FICN (fake currency) seizures** that are defeating manual detection.
- These are industrialised operations spanning multiple states, run by organised networks rather than individual actors.

What investigators lack isn't evidence after the fact — it is intelligence before mass victimisation, and tools that reveal the whole network rather than one flagged transaction at a time.

Our Solution

CyberInvestigator addresses this gap through seven core capabilities:

1. Detects coordinated fraud rings using graph community detection (Louvain) over shared devices, UPI handles, and money flow.
2. Identifies the ringleader via betweenness centrality on the directed money-flow graph — the entity through which funds are funnelled.
3. Scores every entity explainably by fusing a transparent rule engine, unsupervised anomaly detection (IsolationForest), and a supervised model (RandomForest) into a single score, accompanied by human-readable reasons for every flag.
4. Reconstructs the lead-time story — precisely when a ring became detectable versus how many victims were defrauded afterward.
5. Maps cross-jurisdiction spread using DBSCAN geo-clustering alongside a live map and rotating-globe visualisation.
6. Answers investigator questions live via a tool-calling LLM agent (OpenRouter) that must cite specific entity and transaction IDs for every claim, with a deterministic offline fallback when no API key is configured.
7. Seals court-admissible evidence — one click produces a SHA-256 hash-chained Intelligence Package that is tamper-evident and fully auditable.

System Architecture

The platform follows a linear, auditable data flow:

Synthetic Dataset (persons, accounts, UPI, devices, IP, transactions)

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Entity Resolution & Features | **Scoring Engine** (Rule Engine + IsolationForest + RandomForest) | **Graph Engine** (NetworkX: Louvain, Betweenness, WCC, shortest-path, DBSCAN)



FastAPI + Hash-Chained Ledger (with optional Neo4j Aura for graph storage)



React Command Centre (graph / map / globe / evidence UI) and **Tool-Calling Agent** (OpenRouter)

Key design decision: detection runs entirely in-process using NetworkX, pandas, and scikit-learn — no external graph database is required. Neo4j Aura’s free/standard tier does not ship the Graph Data Science library, so rather than depend on it, every graph algorithm runs directly against the data in Python. Neo4j and the LLM agent are wired in as optional enhancers at the edges of the system; the platform runs and detects fraud rings correctly with zero external services.

Technology Stack

Layer	Choice
Frontend	React 18, TypeScript, Vite, Tailwind CSS v4
Graph Visualisation	Cytoscape.js
Map	Leaflet + OpenStreetMap
Backend	FastAPI (Python), modular monolith
Detection Engine	NetworkX, pandas, scikit-learn
ML Models	RandomForest + IsolationForest, layered under a transparent rule engine
Graph Database (optional)	Neo4j Aura
LLM	OpenRouter — model-agnostic, tool-calling
Ledger	SHA-256 hash chain in a flat JSON file — zero extra infrastructure

Validated Results

Result	Detail
Ring detection purity	Louvain community detection recovers the planted fraud ring at 100% purity
Model performance (PR-AUC)	0.77 — a realistic, non-inflated number; an account-age data leak was identified and fixed during testing
Entities analysed	1,500
Fraud rings detected	3
Entities flagged high-risk	33
Amount at risk identified	₹1.76 crore
Lead-time example	Ring became detectable at victim #3; 62 further victim payments occurred afterward

Key Differentiators

- Explainable AI with human-readable reasons for every flag — not a black-box score.
- A tool-calling investigation agent that reasons over live evidence and must cite entity/transaction IDs, rather than summarising a static document.
- Tamper-evident, hash-chained intelligence packages purpose-built for legal admissibility.
- Cross-jurisdiction visualisation showing fraud rings that span multiple states — visibility no single police station has in isolation.
- A lead-time metric that quantifies prevented harm, not just detection accuracy.

Scope Decisions

The hackathon brief outlines five illustrative tracks: Digital Arrest Scam Detection, Counterfeit Currency Identification, Fraud Network Graph Intelligence, Geospatial Crime Pattern Intelligence, and Citizen Fraud Shield. Rather than build shallow versions of all five within the available time, the team deliberately went deep on Fraud Network Graph Intelligence as the primary track, with Geospatial Crime Pattern Intelligence as a secondary, complementary track. This was a strategic choice to maximise technical depth and judge-visible rigor within the build timeframe, not a limitation of scope or ambition.

Judging Criteria Alignment

Criterion	Weight	How This Project Addresses It
Innovation	25%	Tool-calling, evidence-citing agent combined with hash-chained, court-admissible intelligence packages
Business Impact	25%	The lead-time metric — a quantified, victims-prevented-if-acted-earlier number
Technical Excellence	20%	Real graph algorithms (Louvain, betweenness, weakly connected components, shortest-path, DBSCAN) and explainable fused scoring, verified against actual data rather than only claimed
Scalability	15%	Modular monolith with a clear microservice-split path; batch pipeline today, queue-based ingestion documented as the production path for national volume
User Experience	15%	One coherent command centre combining graph, map, agent, and evidence in a single investigative flow

Roadmap

- Swap RandomForest for XGBoost with SHAP for per-feature attribution charts in the entity view.
- Wire the graph view to optionally read live from Neo4j when GDS/Aura is available.
- Introduce a citizen-facing lightweight fraud-check widget reusing the same backend.
- Adopt queue-based ingestion to support national-scale data volume.

Disclaimer

Built for a national AI hackathon on entirely synthetic data. Not connected to any real financial system, law-enforcement database, or citizen data. Not intended for production use as-is.

Repository & Resources

GitHub repository: <https://github.com/AayushG10/cyber-intelligence-knowledge-base-investigator>

The repository includes the full source code, an editable architecture diagram (Excalidraw format), a presentation deck, a demo video recording script, and this submission document.